

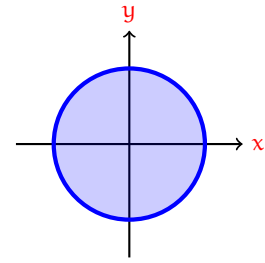
Example 1. If they exist, find the maximum values of

$$f(x, y) = xy$$

on the disk $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$.

Solution. First we address existence of extremizers.

Observe that $f(x, y) = xy$ is **continuous** because



Next observe that the disk D is **bounded** because

and that the disk D is **closed** because

Being closed and bounded means D is:

By the extreme value theorem, a continuous function on a compact feasible set is

Now let's locate them.

First we address critical points in the interior by setting:

We observe that this point is:

Thus we record:

Next let's address the edge $x^2 + y^2 = 1$.

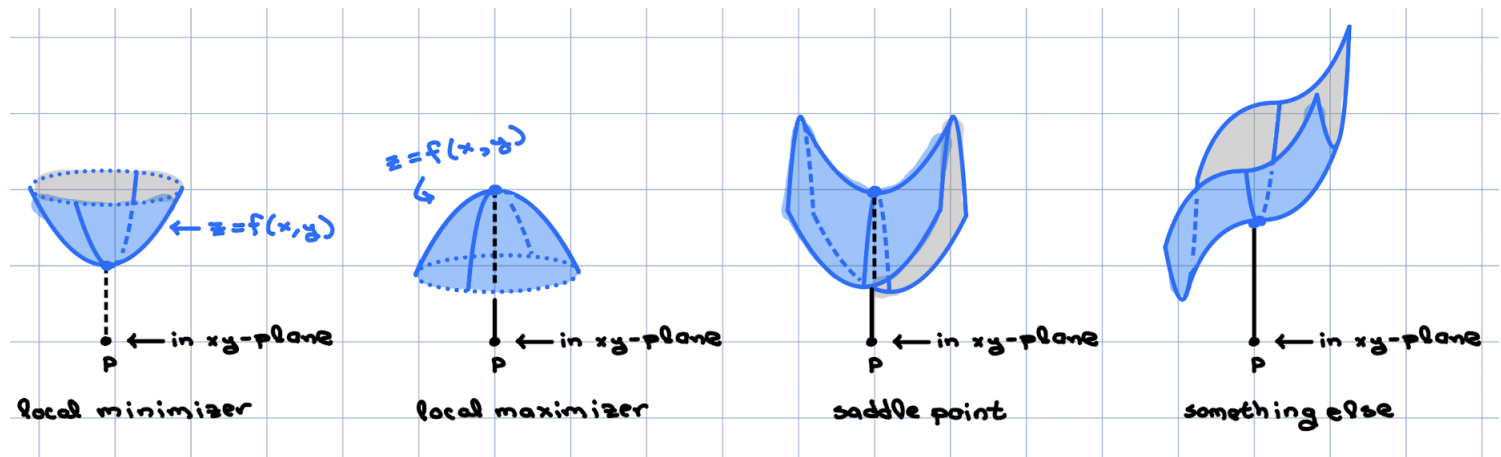
We can parametrize the edge using the polar angle as

We use our parameter to write f as a 1-variable function along the edge.

We also use trig identity $\sin 2\theta =$

By the nature of sine, the maximum along the edge occurs when

A. **Second Derivative Test.** We investigate behavior at critical points.



Concavity is related to second derivatives.

We encode the second derivatives of $f(x, y)$ in a matrix called the **Hessian**.

The determinant of the Hessian is $D =$

If $D < 0$ at a critical point then concavity at the point is

and thus the critical point is called a:

If $D > 0$ at a critical point then concavity at the point is

In this case we compute the **trace** of the Hessian, $T =$

If $D > 0$ and $T > 0$ at a critical point then concavity at the point is

and thus the critical point is called a:

If $D > 0$ and $T < 0$ at a critical point then concavity at the point is

and thus the critical point is called a:

We call this classification based on the Hessian the **Second Derivative Test**.

The remaining type of critical point is where:

In this case the second derivative test is:

Example 2. Find and classify all critical points of

$$f(x, y) = 2x^3 + 3x^2 + 12xy + 3y^2 - 6y$$

Solution. First we locate critical points by setting:

Next we calculate the Hessian.

We evaluate the Hessian at one critical point and compute D and T as needed.

We evaluate the Hessian at the other point and compute D and T as needed.

Example 3. Find and classify all critical points of

$$f(x, y) = xy + \frac{1}{x} + \frac{1}{y}$$

Solution. First we locate critical points by setting:

Next we calculate the Hessian.

We evaluate the Hessian at the critical point and compute **D** and **T** as needed.